

Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations

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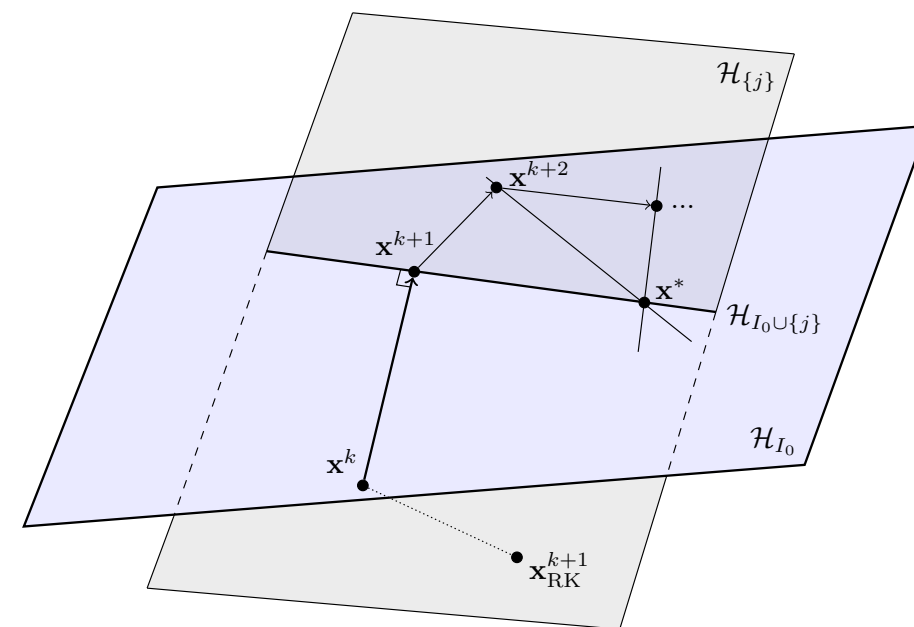


Subspace-constrained sketch-and-project

Suppose we constrain the dynamics of the *sketch-and-project* method for solving $\mathbf{Ax} = \mathbf{b}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, within the (affine) subspace

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{QA}\mathbf{x} = \mathbf{Qb}\}$$

defined by $\mathbf{Q} \in \mathbb{R}^{d \times m}$ with $d < m$.



First, solve $\mathbf{QA}\mathbf{x}^0 = \mathbf{Qb}$ for the initial iterate $\mathbf{x}^0$. Then, iterate:

$$\mathbf{x}^{k+1} = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{x} - \mathbf{x}^k\|_{\mathbf{B}} \quad \text{such that} \quad \mathbf{S}^k \mathbf{Ax} = \mathbf{S}^k \mathbf{b}, \quad \mathbf{QA}\mathbf{x} = \mathbf{Qb},$$

where $\|\mathbf{z}\|_{\mathbf{B}} = \sqrt{\mathbf{z}^T \mathbf{B} \mathbf{z}}$ and $\mathbf{S}^k \sim \mathcal{D}$ are random sketching matrices.

Upshot: Same updates as usual sketch-and-project with $\mathbf{AP}_{\mathbf{B}}$ in place of $\mathbf{A}$, where $\mathbf{P}_{\mathbf{B}}$ is the oblique projector onto $\text{Null}(\mathbf{QA})$ w.r.t. $\|\cdot\|_{\mathbf{B}}$.

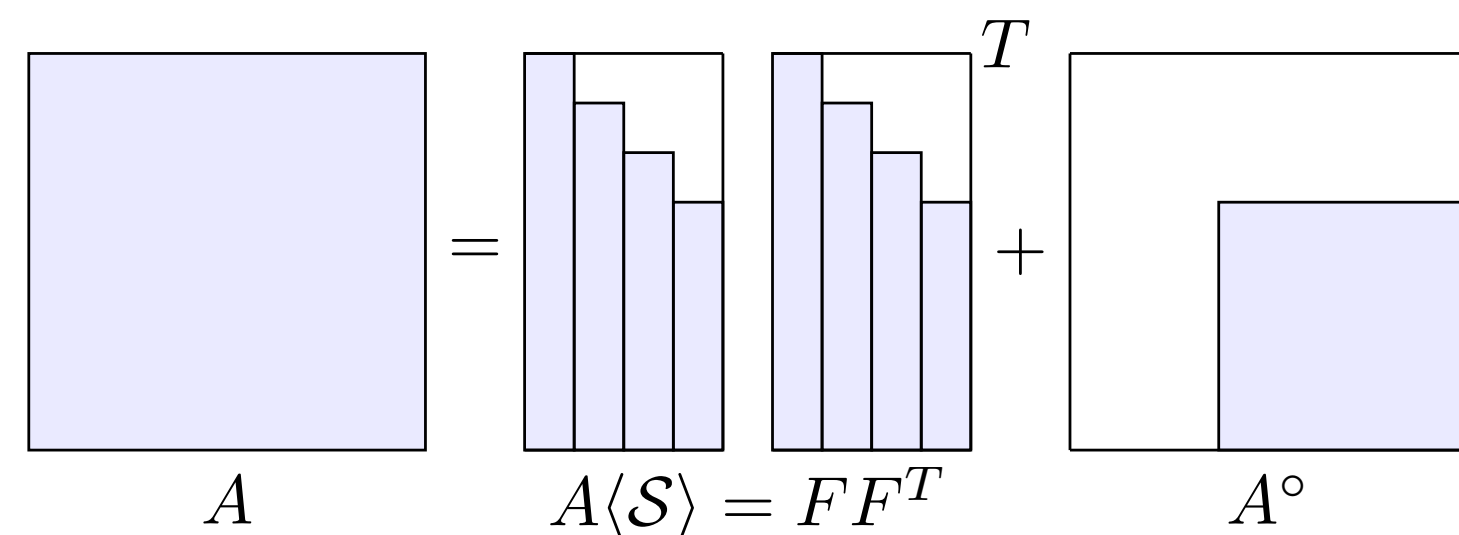
Solving positive semidefinite linear systems

Specifically, consider solving $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is psd.

Example: *Kernel ridge regression* is equivalent to solving $(\mathbf{K} + \lambda \mathbf{I})\mathbf{x} = \mathbf{y}$, where $\mathbf{K}$ is a kernel matrix measuring the similarity between n pairs of data points and $\lambda > 0$ is a regularization parameter.

Randomized block coordinate descent: choose $\mathbf{B} = \mathbf{A}$ and sparse sketches $\mathbf{S} = \mathbf{e}_{\mathcal{J}}^T$ corresponding to a block $\mathcal{J} \subseteq \{1, 2, \dots, n\}$ of size ℓ .

Subspace constraint: Choose $\mathbf{Q} = \mathbf{e}_{\mathcal{S}}^T$ also aligned with coordinate basis. In particular, want to find a set $\mathcal{S} \subseteq \{1, 2, \dots, n\}$ of d coordinates corresponding to a *good low-rank Nyström approximation* $\mathbf{A}\langle \mathcal{S} \rangle$ of $\mathbf{A}$ such that the *residual matrix* $\mathbf{A}^\circ = \mathbf{A} - \mathbf{A}\langle \mathcal{S} \rangle \approx 0$.



A good option is the **RPCholesky** algorithm (Chen et al. [2025]). This computes a near-optimal rank- d psd approximation using $O(nd)$ memory and $O(nd^2)$ arithmetic operations via **adaptive diagonal sampling**.

Subspace-constrained randomized coordinate descent (SC-RCD) method

For solving $\mathbf{Ax} = \mathbf{b}$ with psd $\mathbf{A} \in \mathbb{R}^{n \times n}$:

1. Compute rank- d Nyström approx. $\mathbf{A}\langle \mathcal{S} \rangle = \mathbf{FF}^T$ (RPCholesky)
2. **Initialize:** $\mathbf{C} = (\mathbf{F}_{\mathcal{S},:})^{-T} \mathbf{F}^T$, $\mathbf{x}^0 = -\mathbf{e}_{\mathcal{S}}^T (\mathbf{F}_{\mathcal{S},:})^{-T} (\mathbf{F}_{\mathcal{S},:})^{-1} \mathbf{b}$
3. **Iterate:** Sample $\mathcal{J} = \{j_1, \dots, j_\ell\} \sim \text{i.i.d.} \text{diag}(\mathbf{A}^\circ) / \text{tr}(\mathbf{A}^\circ)$, compute

$$\boldsymbol{\alpha}^k = (\mathbf{A}_{\mathcal{J},\mathcal{J}} - \mathbf{F}_{\mathcal{J},:} (\mathbf{F}_{\mathcal{J},:})^T)^\dagger ((\mathbf{A}_{:, \mathcal{J}})^T \mathbf{x}^k - \mathbf{b}_{\mathcal{J}}) \in \mathbb{R}^\ell,$$

and update $d + \ell$ coordinates of $\mathbf{x}^k$ in $\mathcal{S} \cup \mathcal{J}$:

$$\mathbf{x}_{\mathcal{J}}^{k+1} = \mathbf{x}_{\mathcal{J}}^k - \boldsymbol{\alpha}^k, \quad \mathbf{x}_{\mathcal{S}}^{k+1} = \mathbf{x}_{\mathcal{S}}^k + \mathbf{C}_{:, \mathcal{J}} \boldsymbol{\alpha}^k.$$

Convergence results for SC-RCD

Theorem 1 (Convergence rate). Let $\{\mathbf{x}^k\}_{k \geq 0}$ be the iterates of SC-RCD with subspace constraint $\mathcal{S}$ corresponding to residual matrix $\mathbf{A}^\circ = \mathbf{A} - \mathbf{A}\langle \mathcal{S} \rangle$, and block size ℓ with i.i.d. diagonal sampling. Then, for any solution $\mathbf{x}^*$,

$$\mathbb{E} \|\mathbf{x}^k - \mathbf{x}^*\|_{\mathbf{A}}^2 \leq \left(1 - \frac{\lambda_{\min}^+(\mathbf{A}^\circ)}{\text{tr}(\mathbf{A}^\circ)}\right)^{k\ell} \cdot \|\mathbf{x}^0 - \mathbf{x}^*\|_{\mathbf{A}}^2.$$

Note: Iteration complexity improves at least linearly in the block size ℓ (more generally, holds for sketch-and-project when $\mathbf{S}^k$ has i.i.d. rows).

Given an approximation rank d , let r be the largest integer satisfying $d \geq r + r \log(1/\eta_r)$, where $\eta_r = \sum_{i > r} \lambda_i(\mathbf{A}) / \sum_{i=1}^n \lambda_i(\mathbf{A})$. Define the **normalized tail condition number** of $\mathbf{A}$ by

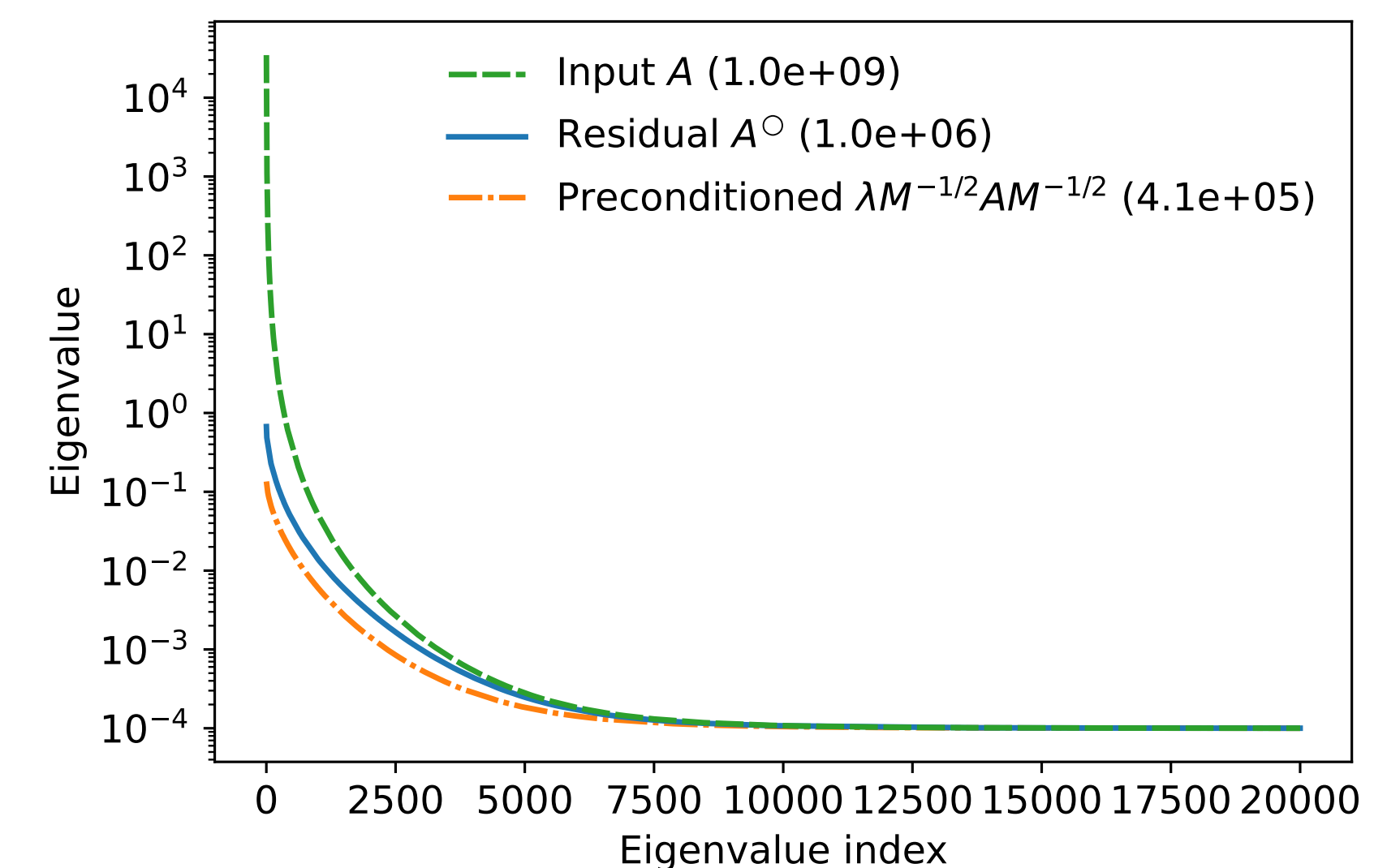
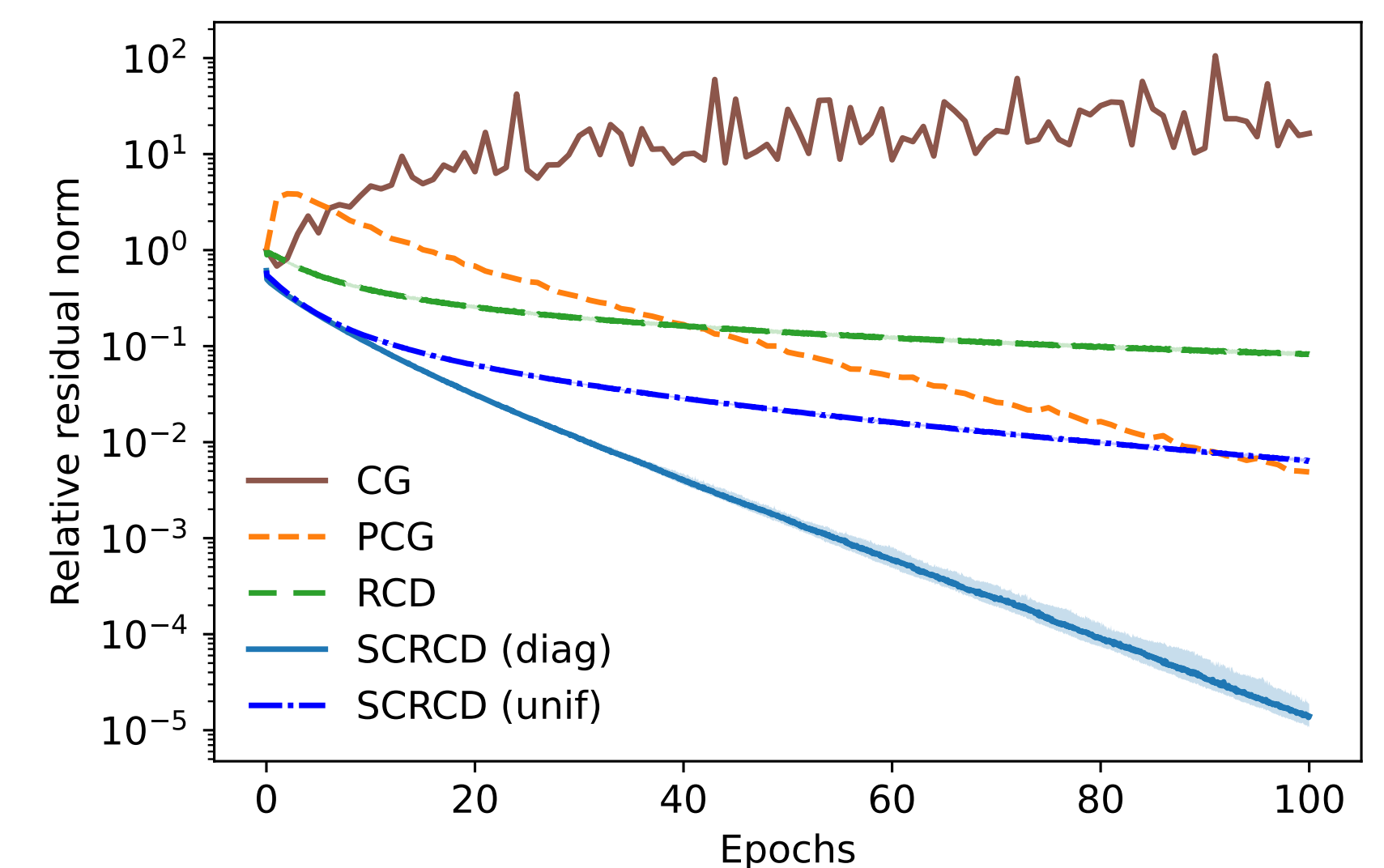
$$\bar{\kappa}_r(\mathbf{A}) := \frac{1}{n - r} \sum_{i > r} \frac{\lambda_i(\mathbf{A})}{\lambda_{\min}^+(\mathbf{A})}.$$

Theorem 2 (Complexity). The SC-RCD method with approximation rank d and block size $\ell = d$ computes $\mathbf{x}^k$ satisfying $\mathbb{E} \|\mathbf{x}^k - \mathbf{x}^*\|_{\mathbf{A}}^2 \leq \varepsilon \|\mathbf{x}^0 - \mathbf{x}^*\|_{\mathbf{A}}^2$ using $O(nd)$ storage and $O((n^2 + nd^2) \cdot \bar{\kappa}_r(\mathbf{A}) \log(1/\varepsilon))$ arithmetic operations.

Note: Under a mild assumption that $\mathbf{A}$ is not extremely ill-conditioned, r can be taken as large as $r = O(d/\log n)$. In particular, if $\bar{\kappa}_r(\mathbf{A}) = O(1)$ for some $r = O(\sqrt{n}/\log n)$, then SC-RCD with $d = \ell = O(\sqrt{n})$ can compute an ε -approximate solution using $O(n^2 \log(1/\varepsilon))$ operations.

Numerical results

Consider solving a kernel ridge regression problem $(\mathbf{K} + \lambda \mathbf{I})\mathbf{x} = \mathbf{y}$ with $n = 100,000$ and tiny regularization $\lambda = 10^{-9}n$ from a particle physics dataset. (Top) SC-RCD with $d = \ell = 1,000$ is an effective solver that does not require the kernel matrix $\mathbf{K}$ to be stored in memory ($\approx 80\text{GB}$). A *preconditioned conjugate gradient method* (PCG) based on RPCholesky (Díaz et al. [2024]) is also included for comparison. (Bottom) $\mathbf{A} = \mathbf{K} + \lambda \mathbf{I}$ exhibits rapid spectral decay.



References

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